

Synthetic Historical Well Drilling Report

SYNTHETIC DEVELOPMENT DOCUMENT. This report is generated from the NWIS MVP synthetic database for demonstration and testing. It is NOT a real Oil India Limited drilling report and must not be treated as operational or confidential OIL material.

Well identification

Well ID: W020. Field: Field-D. Well type: Exploration. Status: Completed. Total depth: 3890.0 m. Surface location (synthetic): latitude 23.888932, longitude 87.155992.

Formation intervals

Formation-A: 0.0-979.9 m. Lithology: Sandstone. Pressure: 37.81. Temperature: 45.09.

Formation-B: 969.2-1962.3 m. Lithology: Shale. Pressure: 44.53. Temperature: 56.51.

Formation-C: 1957.4-2937.0 m. Lithology: Limestone. Pressure: 53.56. Temperature: 70.64.

Formation-D: 2912.7-3890.0 m. Lithology: Dolomite. Pressure: 61.62. Temperature: 90.97.

Drilling observations

This well has 150 synthetic drilling-log samples in the NWIS database. The following observations summarize those records by formation interval. No additional drilling events are inferred beyond the stored logs and events table.

Drilling observations in Formation-A (0.0-979.9 m, lithology Sandstone): 22 synthetic log samples. Mean ROP 30.27 m/h, mean WOB 11.47, mean RPM 115.57, mean torque 14.85, mean standpipe pressure 2387.06, mean mud density 1.09. Non-normal event_label rows in this interval: 0.

Drilling observations in Formation-B (969.2-1962.3 m, lithology Shale): 44 synthetic log samples. Mean ROP 27.14 m/h, mean WOB 12.04, mean RPM 116.84, mean torque 16.30, mean standpipe pressure 2459.48, mean mud density 1.11. Non-normal event_label rows in this interval: 0.

Drilling observations in Formation-C (1957.4-2937.0 m, lithology Limestone): 43 synthetic log samples. Mean ROP 25.22 m/h, mean WOB 13.00, mean RPM 114.62, mean torque 17.37, mean standpipe pressure 2577.00, mean mud density 1.13. Non-normal event_label rows in this interval: 0.

Drilling observations in Formation-D (2912.7-3890.0 m, lithology Dolomite): 43 synthetic log samples. Mean ROP 19.13 m/h, mean WOB 13.81, mean RPM 113.91, mean torque 20.70, mean standpipe pressure 2631.81, mean mud density 1.14. Non-normal event_label rows in this interval: 15.

Historical events

3 historical event record(s) are stored for W020. Each item below is copied from the synthetic events table.

Historical event at 3526.0 m in Formation-D. Event type: MUD_LOSS. Severity: HIGH. Observation: Mud volume loss was observed during drilling.. Cause: Losses into a permeable or fractured formation interval.. Mitigation: Reduced pump rate and applied controlled lost-circulation treatment.. Outcome: Mud losses stabilized and drilling continued..

Historical event at 3639.7 m in Formation-D. Event type: STUCK_PIPE. Severity: HIGH. Observation: Drill string experienced elevated resistance and became temporarily stuck.. Cause: Increasing torque and declining rate of penetration in a mechanically unstable interval.. Mitigation: Circulation, controlled WOB reduction, and reciprocation of the drill string.. Outcome: Drilling

resumed after controlled freeing operation..

Historical event at 3662.5 m in Formation-D. Event type: HIGH_TORQUE. Severity: MEDIUM. Observation: Torque increased significantly while drilling through the interval.. Cause: Increased formation resistance and poor hole-cleaning conditions.. Mitigation: Adjusted WOB/RPM and improved circulation parameters.. Outcome: Torque returned toward the normal operating range..

Source statement

All well, formation, drilling-log, and event values in this PDF were copied from the local NWIS synthetic PostgreSQL database. The document does not add wells, depths, causes, mitigations, or outcomes that are absent from those tables.